Combinatorial Commutative Algebra: Style guidelines for authors

Ayşe First   Elizabeth Second   Séraphine Third   Tomoaki Fourth
Lina Q. Fifth   Zoë L. Sixth

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Abstract

This note illustrates the main features of the style used for articles in the journal Combinatorial Commutative Algebra. It also serves a style guideline for authors preparing publications for said journal.

Subject Classifications: 05E40, 13F70

Keywords: combinatorics, algebra, commutative ring

1 Introduction

Combinatorial Commutative Algebra (CCA) is a diamond open-access journal run for and by mathematicians. It publishes high-quality research at the interface of combinatorics and commutative algebra and disseminates articles without fees or access barriers.

Illustrative (non-exhaustive) topics of CCA include: monomial/Stanley–Reisner/edge ideals; toric ideals and semigroup rings; homological invariants and free resolutions; Lefschetz properties; combinatorial topology and f-vectors; Gröbner/initial ideals, determinantal/Schubert contexts; arrangements; tropical geometry and related combinatorial Hodge theory; and applications (e.g., coding theory, algebraic statistics).

2 Preparing articles for CCA

CCA accepts articles prepared using LaTeX. Publication versions will be produced using a recent version of pdfLaTeX. CCA uses the style file `cca.sty` which can be downloaded at INSERT URL. Place this file in the same directory as your article's TeX file to compile.

2.1 Preamble and packages

CCA articles **must** begin with the following two lines

```
\documentclass[12pt]{article}
\usepackage[amsmath]{cca}
```

The option `amsmath` can be omitted if the `amsmath` package is not needed. The packages `geometry`, `amsthm`, `amssymb`, `hyperref`, and `microtype` are automatically loaded. Please do not reload or set any options for these packages. The packages `fullpage`, `authblk`, `babel` are explicitly forbidden, as is any command that would change page or margin sizes.

Note 2.1. To use the `cleveref` package add it as an option to the `cca` package as follows.

```
\usepackage[amsmath,cleveref]{cca}
```

Loading `cleveref` directly may cause it to improperly label environments. Options for the `cleveref` package may be passed as in the following example.

```
\def\cleverefoptions{capitalize,noabbrev}
\usepackage[amsmath,cleveref]{cca}
```

2.2 Mandatory information

All CCA submissions **must** contain the following information.

Title: The title of the article is entered in the preamble using the command `\title{}`. If necessary, a line break can be inserted using `\\` as in `\title{First line\\Second line}`. Footnotes on titles are not allowed.

Authors: For each author, please enter their contact information in the preamble using the following command.

```
\author{Name}{Affiliation\\Address}{email}{website}
```

All arguments are mandatory, but the website argument can be empty. The email address should be entered as text without the `mailto:` URI scheme. Similarly, the website, if any, should be entered as text without the `http(s)://` protocol. Clickable links will be generated upon compilation for both email and website; please ensure these links work as intended. The author names are printed on the first page, while the author contact information is printed at the end by inserting the command `\authorinformation` before `\end{document}`. Footnotes on author names are prohibited, with two exceptions (see Note 2.2).

Dates: Authors may provide submission and acceptance dates in the preamble via

```
\dateline{<Submitted>}{<Accepted>}{TBD};
```

production will fill in the publication date. Dates must be entered in the format "Month Day, Year", with month abbreviated to the first three letters (e.g.: Jan, Dec).

Abstract: A brief abstract must be included at the beginning of the submission. The abstract should provide context for the work and list new contributions. It should be self-contained and without references to the bibliography or other parts of the submission. The use of TeX code in the abstract is not recommended as the resulting text may not visualize correctly in a web browser; in particular, user-defined TeX macros must not be used in the abstract. The abstract should be entered in an `abstract` environment placed after the command `\maketitle` at the beginning of the document.

Subject Classifications: Provide at least one 5-digit code from the 2020 Mathematics Subject Classification. Additional codes should be separated by commas.

Keywords: Provide 3 to 6 keywords separated by commas. Keywords should be lowercase, except for proper names. Singular is preferred.

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Note 2.2. Other than copyright information, CCA restricts the use of footnotes, including on title and author names; in particular, the command `\thanks` is forbidden. There are two exceptions.

1. Funding that must be acknowledged on the first page of a publication by explicit requirement of the funding agency. Otherwise, funding should only be mentioned in the Acknowledgment section at the end of the article.
2. Corresponding authors that require acknowledgement on the first page of a publication for institutional purposes. Otherwise, CCA does not recognize a corresponding author in published articles, even though such a role may be a part of the editorial tools used by CCA.

In these cases, enter the required information as a footnote on your name by placing the following code in the preamble:

```
\author[n]{Name}{Affiliation\\Address}{email}{website}
\authornote{n}{Footnote text}
```

where `n` is a positive integer starting at 1. The same footnote can be reused, for example, when two or more authors are funded by the same source. Increase `n` by one for the next footnote.

2.3 Document body

- The document environment should start with `\maketitle` followed by the abstract.
- All sections should be automatically numbered. Do not include a table of contents.
- The default font and font sizes are set by the `cca.sty` file; please do not alter them.
- Accented letters in names and text, such as Gröbner or Erdős, can be input directly as Unicode characters instead of LaTeX commands. For example, á can be used instead of `\'a`.
- The `cca.sty` file defines the following environments: theorem, lemma, corollary, proposition, fact, observation, claim, note, definition, example, conjecture, open, problem, question, remark. The environments are numbered sequentially within sections.
- Equations environments are also numbered within sections. If an equation is not referred to in the rest of the article, the numbering should be suppressed.
- Figures and tables should always be included as floats within a figure and table environment respectively. All floats must be labeled, captioned, and are automatically numbered within sections. As the position of a float may change during production, please refer to it by label (e.g.: Figure 3.1) rather than using expressions such as the figure/table below/above.

- In the interest of producing high quality figures, we strongly recommend creating images using the PGF/TikZ package and including code directly into your LaTeX file. As an alternative, authors may submit image files in PDF or EPS format, preferably containing vector graphics.

2.4 Bibliographic references

All CCA articles must include a complete and up-to-date list of bibliographic references. Such information can usually be retrieved from databases like Mathscinet and zbMath; the latter can also be used for arXiv preprints. All references appearing in the bibliography must be cited in the text. Please make citations as explicit as possible by including an environment number (e.g.: Theorem 2.1(a), Example IV), a (sub)section number, or a page number. Upon compilation, backlinks from the references to the text will be automatically inserted.

CCA recommends using BibTeX with the `abbrv` bibliography style, then copying and pasting the contents of the compiled `.bbl` file into the main source file. If a dedicated `.bib` is used for the submission, then it can be uploaded along with the source file. If not using BibTeX, please use `thebibliography` environment and format your references as close as possible to the aforementioned `abbrv` style.

Whenever possible, CCA suggests adding arXiv and DOI links after each reference inside `thebibliography` environment. For arXiv papers, use `\arxiv{YYMM.number}` or the appropriate identifier. For published works, use `\doi{number}`. See this document and its source for examples.

3 Sample content

Definition 3.1. A commutative ring is called *amazing* if it is isomorphic to the edge ring of an amazing graph.

Example 3.2. The complete graph on five vertices (see Figure 3.1) is amazing. The fictional index of the graph can be found by solving for f in

$$v - e + f = 2, \tag{3.1}$$

where v is the number of vertices and e is the number of edges.

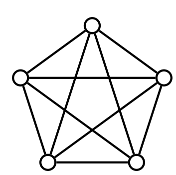

Figure 3.1: A complete graph on five vertices.

Theorem 3.3. *Every amazing ring is excellent.*

Proof. It will be enough to show that an amazing graph is astounding since the edge ring of an astounding graph is excellent as shown in ... □

The reader will find many interesting examples of the interactions between combinatorics and commutative algebra in the vast literature on the subject which includes works such as [1, 2, 3, 4].

Acknowledgements

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References

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Author Information

Ayşe First (first@email.edu, example.org/~first)
Department of Mathematics, University of the North
Coldtown, YT, Canada

Elizabeth Second (second@email.edu, example.org/~second)
Department of Mathematics and Statistics, Universidade do Sul
Rio de Janeiro, Brazil

Séraphine Third (third@email.edu)
Department of Combinatorics and Optimization, Central College
Centertown, OH, USA

Tomoaki Fourth (fourth@email.edu, example.org/~fourth)
Mathematics Department, Mountain Regional University
Tokyo, Japan

Lina Q. Fifth (fifth@email.edu, example.org/~fifth)
School of Mathematics, Seaside College
Cape Town, South Africa

Zoë L. Sixth (sixth@email.edu)
Fictional Department, Imaginary University
Realtown, United Kingdom

Corrigendum – submitted Oct 11, 2026

Author(s) may submit a corrigendum if:

- the main result(s) remain true; and
- author(s) can provide a corrected or replacement proof, or a precise amendment (e.g., added hypothesis, corrected bound) that preserves the paper's core contributions.

If the error makes a central statement false, substantially weakens the results beyond the paper's scope, or requires extensive new theory, the editors may instead pursue a retraction or invite a new, standalone submission.

Communicating a Corrigendum to CCA

To request a corrigendum, the author(s) should start a new submission including the following information.

- A new section appended to the original paper by placing

  ```
  \corrigendum{Month Day, Year}
  ```

  immediately after `\authorinformation` in the accepted manuscript's source file (using cca style). The date is the day author(s) submit the corrigendum to the managing editors. The corrigendum section must include an opening paragraph explaining:

  (i) the issue,
  (ii) why a correction/replacement proof is needed,
  (iii) which results are impacted by the changes.
- For each correction: make sure to include the precise *location* in the published version (section/theorem/equation/figure; page/line if helpful) and the *corrected text/formula*.
- If supplying a new proof for a theorem, restate the theorem, and provide a self-contained proof that introduces no new results and no renumbering (unless the fix requires).

Review and outcome

Corrigenda that include a replacement proof or change mathematical content will be externally checked (preferably by the original referee(s) or a subject expert), with the review focused on the corrected portions unless broader interaction is required.

Upon a positive check, the corrigendum is appended to the published PDF and the article page notes "A corrigendum was added on [date]"; the DOI and record remain unchanged.